Ritesh Somashekar

Data Engineer | AWS, Snowflake & Python

Fairfax, VA | (571) 245-7770 | ritesh.chandra33@gmail.com | LinkedIn | GitHub

PROFESSIONAL SUMMARY

- **Data Engineer** with **6+ years** building **ETL/ELT pipelines** and cloud data platforms on **AWS** and **Snowflake** for healthcare, industrial IoT, and research; pursuing **AWS Certified Data Engineer - Associate**.
- Hands-on with **AWS Glue, S3, Lambda, EventBridge, CloudWatch, Athena, SQS,** and **SNS** plus **Apache Airflow**, delivering **99.9% pipeline reliability** with logging, lineage tracking, and automated data quality checks.
- Skilled in **Python, SQL, PySpark, Snowflake, Databricks,** and **Delta Lake**, with **HIPAA-compliant** data handling and published research at **ACM WSDM '26** and **IEEE PST 2026**.

TECHNICAL SKILLS

AWS Data & Serverless: AWS Glue, **Amazon S3**, **AWS Lambda**, **Amazon EventBridge** (Scheduler, Rules), Amazon Athena, Amazon SQS, Amazon SNS, Amazon RDS, API Gateway

Logging & Monitoring: Amazon CloudWatch (Logs), S3 Log Archival, Pipeline Monitoring & Alerting, Data Lineage, Audit Logging

Data Warehousing & Lakes: Snowflake, Databricks, Delta Lake, Azure Data Lake, Star Schema, Data Modeling, Data Quality, Data Validation

Pipelines & Orchestration: ETL/ELT, **Apache Airflow**, Apache Spark, **PySpark**, Kafka, Azure Data Factory, dbt, Hadoop, Hive, Informatica PowerCenter

Programming & Databases: Python, **SQL**, R, Scala, Bash, SQL Server, MySQL, PostgreSQL, MongoDB, ChromaDB

DevOps & Tools: Git, GitHub, CI/CD, Docker, Kubernetes, DVC, FastAPI, REST APIs, Jira, Postman, Tableau, Power BI

EDUCATION

Master's in Data Analytics and Engineering | George Mason University, Fairfax, VA, USA

Bachelor's in Information Science and Engineering | Visvesvaraya Technological University, Bengaluru, Karnataka, India

CERTIFICATIONS

AWS Certified Data Engineer - Associate (DEA-C01) | In Progress

Study areas: AWS Glue, S3, Lambda, EC2, EventBridge, CloudWatch, CloudTrail, VPC Flow Logs, Redshift, Kinesis

PROFESSIONAL EXPERIENCE

George Mason University | ***Graduate Research Assistant*** **Fairfax, VA | August 2024 - Present**

- To automate service-ticket triage for a cybersecurity partner, built a **serverless AWS pipeline** of **Lambda** functions triggered by **EventBridge Scheduler** rules and **S3 events**, reaching **93% automation accuracy** and **70%** faster triage.
- To make every pipeline run traceable, streamed Lambda logs to **Amazon CloudWatch**, archived run logs in **S3**, and sent SLA breach alerts through **SNS**, cutting operational risk by **40%** with audit-ready records.
- To scale ticket processing, designed an **event-driven backend** chaining Lambda stages through **S3 event triggers** and **SQS** messaging, with **Jira REST API** ingestion into S3-staged JSON and CSV data, improving scalability by **50%**.
- To give 9-1-1 training AI trustworthy data, led the **data pipeline** for **SHIELD** and **MATE-911** in **Python**, turning unstructured call records into **160** validated scenarios and cutting hallucinations from **13.0% to 3.0%**.
- To stop bad records reaching production, built **data quality gates** (completeness, schema, ontology, and 85% similarity checks) before loading into **ChromaDB**, lowering hallucinations from **9.5% to 3.0%**.
- To prepare research data at scale, built pipelines with **Python, SQL, Pandas, AWS S3,** and **DVC** for **30,000 documents** (cleaning, deduplication, metadata tagging), improving entity-recognition F1 by **18%**.
- To cut data preparation time, standardized **15GB** of structured and unstructured data across six datasets with repeatable transformation and validation, reducing preparation time by **30%**.
- To serve processed data to researchers, deployed **FastAPI** and **Docker** data APIs at **sub-200ms** latency, versioning datasets with **Git** and **DVC** for reproducible workflows.

Carelon Global Solutions (Elevance Health) | ***Data Engineer*** **Bengaluru, India | October 2020 - August 2023**

- To support downstream AI and BI applications, designed and managed **20+ enterprise ETL pipelines** on **AWS**, boosting data throughput by **70%** with high availability.
- To remove manual pipeline runs, automated workflows with **Apache Airflow, Python,** and **AWS (S3, Lambda, Athena)**, achieving **99.9% pipeline reliability** with end-to-end lineage tracking.
- To unify healthcare data under **HIPAA**, built **AWS Glue** and **PySpark** ETL pipelines into **S3** and **Delta Lake**, improving data reliability by **30%**.
- To speed up model training, ingested **400K+ XML/JSON healthcare records** into **Snowflake** and **Databricks**, accelerating training by **60%** and cutting preprocessing time by **70%**.
- To fix inaccurate provider data, built a daily **NPPES** ingestion workflow in **Python** and **SQL**, improving data accuracy by **74%** and cutting processing overhead by **60%**.
- To process growing data volumes, engineered **PySpark** and **Databricks** pipelines handling **500GB+** of structured and unstructured data, reducing processing time by **45%**.
- To give business teams faster reporting, delivered self-service **Tableau** and **Power BI** pipelines on SQL data models, cutting reporting latency by **40%**.
- To reduce manual claims work, redesigned **claims-balancing ETL workflows** and automated medical invoice pipelines, cutting operational overhead by **60%** and manual errors by **70%.**

Honeywell | ***Data Engineer*** **India | January 2019 - September 2020**

- To unify industrial IoT telemetry, built **ETL/ELT pipelines** with **Python, PySpark,** and **Apache Spark**, ingesting data from **8+ source systems** via REST APIs and processing **2M+ records daily** into **Azure Data Lake**.
- To catch pipeline failures early, automated batch workflows with **Azure Data Factory** and **Apache Airflow**, adding validation and monitoring across **15+ daily jobs** and cutting issue detection time to **under 15 minutes**.
- To improve report accuracy, implemented **data quality frameworks** (null, duplicate, schema, and referential integrity checks), reducing downstream reporting errors by **40%** across **3 plant sites**.
- To speed up operational reporting, designed a **star-schema** data warehouse and reusable **dbt** SQL transformations, cutting report turnaround by **60%** across **5 teams** and improving data prep efficiency by **35%**.
- To modernize legacy workloads, migrated **Hive** and **Hadoop/Sqoop** jobs to **Apache Spark** and **Azure Data Services**, improving processing speed and scalability.
- To keep raw data durable and auditable, archived telemetry in **AWS S3** and managed production deployments with **Git, CI/CD,** and Linux shell scripting.

PUBLICATIONS

- R. S. Nayaka, **R. Somashekar**, et al. "SHIELD: A Framework for Online Conversational AI System for Self-Regulated Learning at Emergency Communications Centers." *ACM WSDM '26*. doi:10.1145/3773966.3779411
- R. S. Nayaka, **R. Somashekar**, H. Purohit. "MATE-911: Multimodal AI-powered Training Environment for Next-Generation 911 Dispatcher Preparedness." *IEEE PST 2026*.